

# 07 — Maximum power transfer

Class notes p.13–14, p.16–17, set 2 p.14 · MTE syllabus item 5 (DC and AC) · 2 board question sets + 1 textbook AC problem · needs 06

## Map

1 Power in the load

→

2 The condition  $R_L = R_{Th}$

→

3  $P_{max}$  and the 50 % efficiency

→

4 AC loads: conjugate match

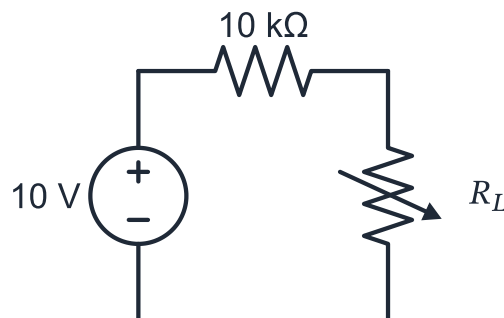
## The questions this file answers

- **Class p.16–17** 10 V source, 10 k $\Omega$ , variable  $R_L$ : for what  $R_L$  is power maximum? Total power from the source? Power in  $R_{Th}$ ? Efficiency? Voltage across the load? How do the load's voltage and power compare with the source's?
- **Class p.17** 10 V source, 5 k $\Omega$ : find R for maximum power, the maximum power, and the efficiency.

## Build

### CLASS P.16

For what value of load resistance will maximum power be delivered? Then: determine the total power delivered by the source.



The board circuit: a 10 V source with 10 k $\Omega$  in series, driving a variable load.

### GUESS FIRST

Make  $R_L$  tiny, then huge. What happens to the load's power at each extreme? So where must the maximum be?

## 1 · Power in the load

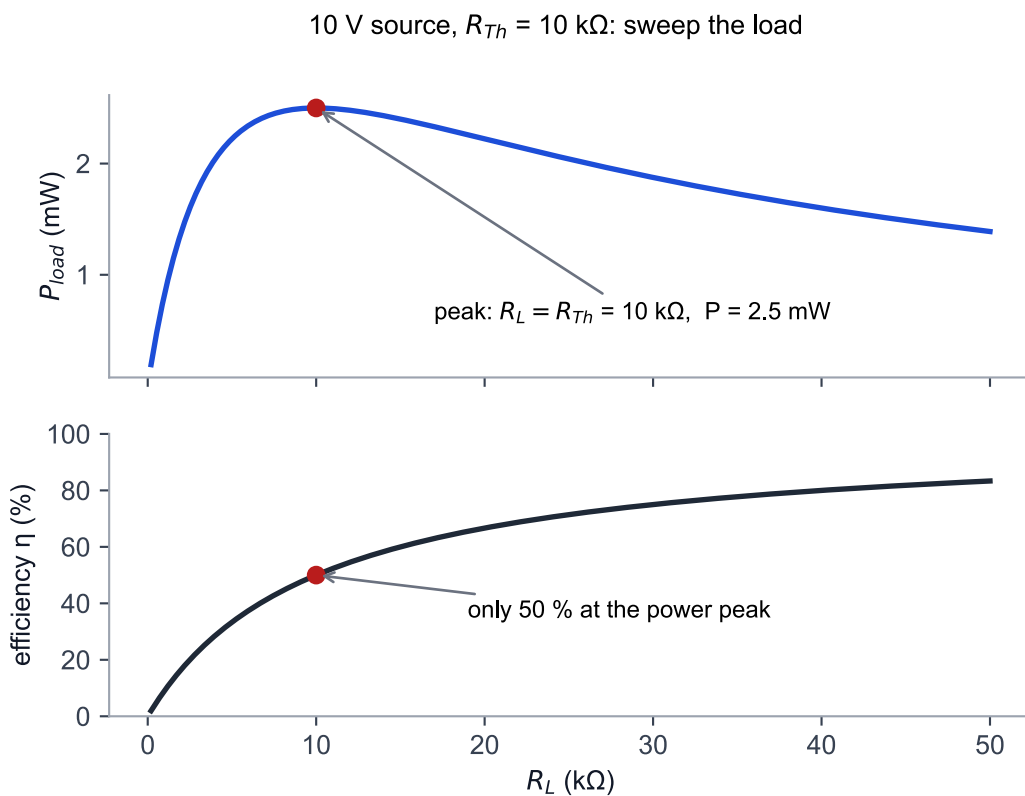
From 06, every linear source network is  $V_{Th}$  in series with  $R_{Th}$ , so this circuit is already in that form. The load current and power:

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} \quad P_L = I_L^2 R_L = \frac{V_{Th}^2 R_L}{(R_{Th} + R_L)^2}$$

### CLASS P.13

The first attempt on the board used  $P_L = V_{Th}^2 / (R_{Th} + R_L)$ , which is wrong; it was crossed out and redone with  $P_L = I_L^2 R_L$  (starred). Use the starred one.

The Guess answer: tiny  $R_L$  gives big current but almost no voltage, so  $P \approx 0$ . Huge  $R_L$  gives big voltage but almost no current, so  $P \approx 0$ . The power must peak somewhere in between:



Top: the load's power peaks exactly at  $R_L = R_{Th}$ . Bottom: efficiency keeps rising with  $R_L$  and is only 50 % at that peak.

## 2 · The condition (class p.13–14 derivation)

Set  $\frac{dP_L}{dR_L} = 0$ . Quotient rule:

$$\frac{dP_L}{dR_L} = V_{Th}^2 \cdot \frac{(R_{Th} + R_L)^2 - R_L \cdot 2(R_{Th} + R_L)}{(R_{Th} + R_L)^4} = 0$$

The numerator must be zero:

$$(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L) = 0 \Rightarrow R_{Th}^2 + 2R_{Th}R_L + R_L^2 - 2R_{Th}R_L - 2R_L^2 = 0 \Rightarrow R_{Th}^2 = R_L^2$$

$$R_L = R_{Th}$$

(The negative root is not a resistance. The curve above shows it is a maximum, not a minimum.)

**Answer, part 1:**  $R_L = R_{Th} = 10 \text{ k}\Omega$ .

### 3 · $P_{max}$ , source power, efficiency

Put  $R_L = R_{Th}$  into  $P_L$ :

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}}$$

For the board circuit, at  $R_L = 10 \text{ k}\Omega$ :

| Quantity                    | Working                                                                       | Value                                        |
|-----------------------------|-------------------------------------------------------------------------------|----------------------------------------------|
| Current                     | 10 V / 20 k $\Omega$                                                          | <b>0.5 mA</b>                                |
| Total power from the source | $V \cdot I = 10 \times 0.5 \text{ mA}$                                        | <b>5 mW</b>                                  |
| Power in the load           | $I^2 R_L = (0.5 \text{ mA})^2 \times 10 \text{ k}\Omega = V_{Th}^2 / 4R_{Th}$ | <b>2.5 mW</b>                                |
| Power in $R_{Th}$           | same current, same resistance                                                 | <b>2.5 mW</b>                                |
| Efficiency $\eta$           | $P_{load} / P_{total} = 2.5 / 5$                                              | <b>50 %</b>                                  |
| Voltage across the load     | $0.5 \text{ mA} \times 10 \text{ k}\Omega$                                    | <b>5 V = <math>V_s/2</math></b>              |
| Load vs source              |                                                                               | <b><math>V_L = V_s/2, P_L = P_s/2</math></b> |

#### CLASS P.16

The notes read " $I \times V = 0.5 \text{ mW}$ " with no visible decimal point. The value is **5 mW**. The 50 % efficiency on p.17 is right either way.

**Maximum power is not maximum efficiency.**  $\eta = \frac{R_L}{R_{Th} + R_L}$  keeps rising as  $R_L$  grows (bottom plot), while  $P_L$  falls. So:

- **communication / signal circuits** match  $R_L = R_{Th}$ : a weak signal needs every bit of power delivered, and 50 % loss is acceptable;
- **power systems** keep  $R_L \gg R_{Th}$ : wasting half the generated power would be unacceptable.

The 50 % refers to the Thévenin equivalent. The power actually lost inside the original, full network can differ.

### CHECK 1

- (a) Why is exactly half the power lost at the match? Use the fact that  $R_{Th}$  and  $R_L$  carry the same current.
- (b) The class p.17 question: 10 V with 5 k $\Omega$ . Find  $R$ ,  $P_{max}$ ,  $\eta$ .

## 4 • AC loads: the conjugate match

### PREDICT

A source has  $Z_{Th} = 3 + j4 \Omega$ . Its reactance  $j4 \Omega$  doesn't absorb power, but it does limit the current. What reactance should the load add?

For impedances (class p.16; set 2 p.14: " $Z_L = Z_{Th}^*$ ,  $*$  = complex conjugate,  $i \rightarrow -i$ ):

$$Z_L = Z_{Th}^* \quad (\text{the complex conjugate: } j \rightarrow -j)$$

- $R_L = R_{Th}$ : the resistance match, as before.
- $X_L = -X_{Th}$ : the load's reactance **cancels** the source's, so the loop is purely resistive and the current is as large as possible.

For the Predict:  $Z_L = 3 - j4 \Omega$ . Then  $P_{max} = \frac{|V_{Th}|^2}{4R_{Th}}$ , with  $V_{Th}$  as an **rms** value (with a peak value it is  $\frac{|V_{Th}|^2}{8R_{Th}}$ ).

### TEXTBOOK

The AC network of 06 §6 has  $V_{Th} = 7.07 \angle -45^\circ$  V (peak) and  $Z_{Th} = 500 - j500 \Omega$ . Find the load for maximum power, and that power.

- **Load:**  $Z_L = Z_{Th}^* = 500 + j500 \Omega$  (500  $\Omega$  with an inductive reactance of 500  $\Omega$ ; at  $\omega = 1000$  that is  $L = 0.5$  H).
- **Loop:**  $Z_{Th} + Z_L = 1000 \Omega$ , purely resistive: the reactances cancel.  $I = \frac{7.07}{1000} = 7.07$  mA peak = 5 mA rms.
- **Power:**  $P = I_{rms}^2 R_L = (5 \text{ mA})^2 \times 500 \Omega = 12.5 \text{ mW}$ . Same by formula:  $V_{rms} = 5 \text{ V}$ ,  $\frac{25}{4 \times 500} = 12.5 \text{ mW}$ .

A computer sweep over every load  $R + jX$  gives its peak at exactly  $R = 500 \Omega$ ,  $X = +500 \Omega$ , 12.5 mW.

## Exam form

**Statement.** A linear network delivers maximum power to a resistive load when  $R_L = R_{Th}$ . The maximum power is  $P_{max} = V_{Th}^2 / (4R_{Th})$ . For an AC network with a complex load, the condition is  $Z_L = Z_{Th}^*$ .

**Derivation.**  $P_L = V_{Th}^2 R_L / (R_{Th} + R_L)^2$ ; set  $dP_L / dR_L = 0 \Rightarrow R_L = R_{Th}$  (step 2).

**At the match.**  $\eta = 50 \%$ ,  $V_L = V_{Th} / 2$ ,  $P_L = P_S / 2$ .

## Traps

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- Using  $P = V_{Th}^2 / (R_{Th} + R_L)$  (the crossed-out attempt on class p.13).
- Saying "maximum power means maximum efficiency".
- In the AC case, matching  $Z_L = Z_{Th}$  instead of the **conjugate**.
- Forgetting to find  $R_{Th}$  first when the source network is more than one resistor: use 06.

## Self-test

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1. Derive  $R_L = R_{Th}$  (three lines).
2.  $V_{Th} = 12\text{ V}$ ,  $R_{Th} = 3\text{ k}\Omega$ . Find  $P_{max}$  and the source power at that point.
3. Why do power companies not operate at maximum power transfer?
4.  $Z_{Th} = 50 - j20\ \Omega$ . Load for maximum power? If  $V_{Th} = 10\text{ V rms}$ , what is  $P_{max}$ ?
5. In the 06 board circuit ( $V_{Th} = 20/3\text{ V}$ ,  $R_{Th} = 8/3\text{ k}\Omega$ ), what  $R_L$  gets maximum power, and how much?

## Answers

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**Check 1.** (a) At the match,  $R_{Th} = R_L$  and both carry the same current, so  $I^2 R_{Th} = I^2 R_L$ : equal power in each, i.e. half of the total. (b)  $R = 5 \text{ k}\Omega$ ;  $P_{max} = 10^2 / (4 \times 5 \text{ k}\Omega) = 5 \text{ mW}$ ;  $\eta = 50 \%$  (the source supplies  $10 \text{ V} \times 1 \text{ mA} = 10 \text{ mW}$ ).

### Self-test.

1.  $P_L = V_{Th}^2 R_L / (R_{Th} + R_L)^2$ ; the numerator of the derivative is  $(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L) = 0$ ; this gives  $R_{Th}^2 = R_L^2$ , so  $R_L = R_{Th}$ .
2.  $P_{max} = 144 / 12 \text{ k} = 12 \text{ mW}$ ;  $I = 12 / 6 \text{ k} = 2 \text{ mA}$ ;  $P_s = 24 \text{ mW}$ .
3. Efficiency would be only 50 %: half of the generated power lost in the source and lines.
4.  $Z_L = 50 + j20 \Omega$ ;  $P_{max} = 100 / (4 \times 50) = 0.5 \text{ W}$ .
5.  $R_L = 8/3 \text{ k}\Omega \approx 2.67 \text{ k}\Omega$ ;  $P_{max} = (20/3)^2 / (4 \times 8/3) = 4.17 \text{ mW}$ .